

Case study 3 - Finding Exoplanets in Archives

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Abstract

The detection and characterization of exoplanets have significantly advanced our understanding of planetary systems beyond the Solar System. This study investigates six stars—GJ 551, HD 22049, HD 115617, GJ 876, WASP 28, and BD+462629—using two primary methods: radial velocity and transit photometry. Radial velocity analysis revealed periodic stellar wobble indicative of planetary presence for GJ 551, HD 22049, and HD 115617, estimating planetary minimum masses of $5.153 M_{\oplus}$, $259.965 M_{\oplus}$, and $5.399 M_{\oplus}$, respectively, along with their orbital parameters. In the transit photometry method, WASP 28 displayed a detectable dip in brightness, confirming a planetary transit, while GJ 876 and BD+462629 did not show such dips in this study. Comparisons with existing literature revealed deviations in mass and eccentricity estimates, attributed to updated datasets and advanced observational techniques. These findings highlight the importance of precision in exoplanet detection methods and provide refined planetary parameters for the studied systems.

1 Introduction

Exoplanets are the celestial objects that orbit a star. The name is derived from extra-solar planets, as traditionally it was believed that only the solar system has planets. In

rare cases exoplanets may also not orbit any star but wander through the space. Usually, it is difficult to spot an exoplanet using telescopes as they are overshadowed by the star's luminosity, hence astronomers observe the effect of exoplanet on its host star and

determine its properties. Some methods used are gravitational microlensing and astrometry, both using the gravitational effect the exoplanet has on the star.

Six stars are used in this case study to find if there are exoplanets orbiting them. GJ 551 also known as Proxima Centauri, cataloged as the closest star to the Sun in the year 1915 by Robert Innes, HD 22049, Epsilon Eridani, known since ancient times, cataloged in 1603 by Johann Bayer, HD 115617, published in 1924 as a part of the Henry Draper Catalog, GJ 876, cataloged in the 20th

century as part of studies on nearby red dwarf stars, WASP 28, found in the 21st century as a part of the WASP (Wide Angle Search for Planets) program, and BD+462629, cataloged in the Bonner Durchmusterung (BD) star catalog, published between 1859 and 1863.

Radial velocity and transit methods are used to analyse the above stars for exoplanet detection. The tools used are DACE, notepad++ for analysing the stars, nasa exoplanet archive for source data and ads for literature review.

2 Methods, Analysis and Results

We are using 6 stars to proceed in this case study. GJ 551, HD 22049, HD 115617, GJ 876, WASP 28, and BD+462629. First 3 through radial velocity method, remaining 3 via transit method.

Radial velocity method is based on the gravitational pull of the planet on the star while orbiting, enabling us to detect the shift in the star's spectrum due to doppler effect. In transit method, the decreased luminosity of the star when the planet crosses over is analysed.

2.1 Radial Velocity Method

In this section, the radial velocity data of three stars are analysed to identify astrometric wobble, with the assumption that the planets are in the same plane as the dusty disc and finally finding the mass of the orbiting object.

2.1.1 GJ 551

With the help of DACE Geneva 2024 radial velocity curve for GJ 551 is observed. Finding the time period to be 11 days from Figure 2, Keplerian is

added and a shift in radial velocity is observed (Figure 1).

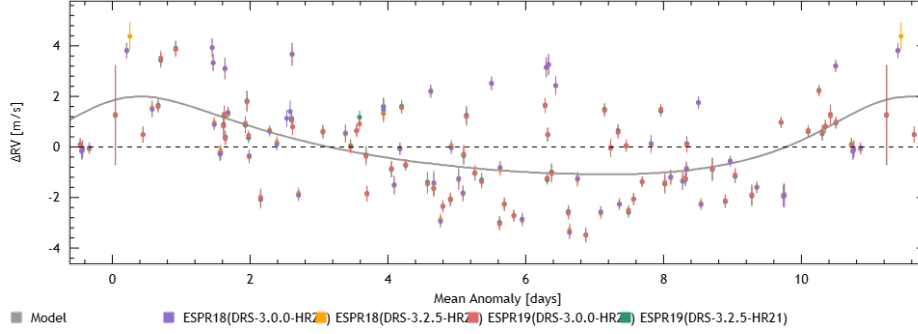


Figure 1: Radial Velocity curve - GJ 551

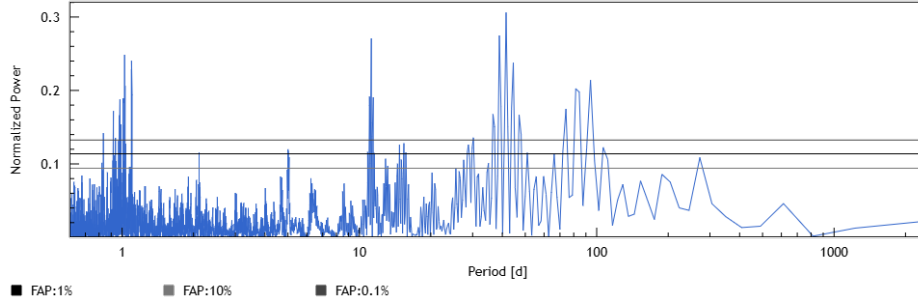


Figure 2: Periodogram - GJ 551

The Keplerian mass for the orbital time period 11.191643 days is $5.153M_{\oplus}$. In addition, the half-amplitude of the signal is measured to be 1.5658 m/s and the eccentricity, 0.3389.

2.1.2 HD 22049

With the help of DACE [\[empty citation\]](#) radial velocity signal for HD 22049 is obtained. Finding the time period to be around 2600 days from the periodogram (Figure 4), Keplerian is added and a shift in radial velocity is observed (Figure 3).

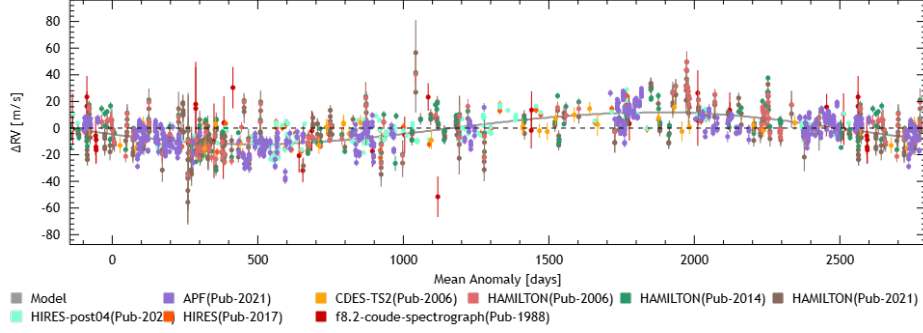


Figure 3: Radial Velocity curve - HD 22049

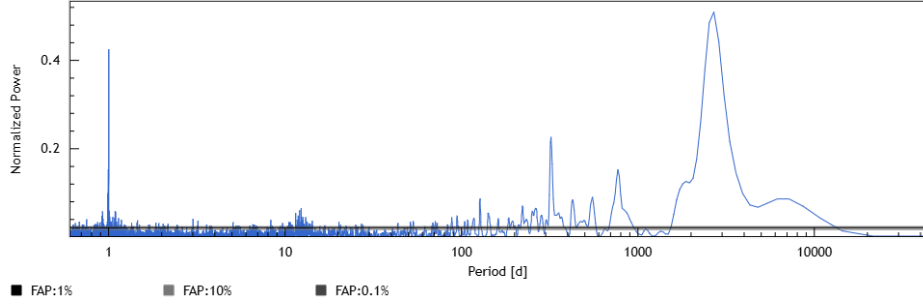


Figure 4: Periodogram - HD 22049

The Keplerian mass for the orbital time period 2649.791 days is $259.965 M_{\oplus}$. Further, the Half-amplitude of the signal is measured to be 12.0173 m/s and the eccentricity, 0.0408.

2.1.3 HD 115617

With the help of DACE [\[empty citation\]](#) radial velocity signal for HD 115617 is obtained. Finding the time period to be 4 days from Figure 6, Keplerian is added and a shift in radial velocity is observed (Figure 5).

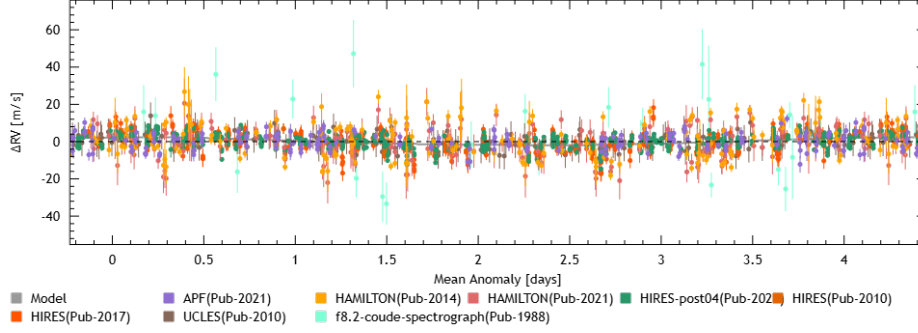


Figure 5: Radial Velocity curve - HD 115617

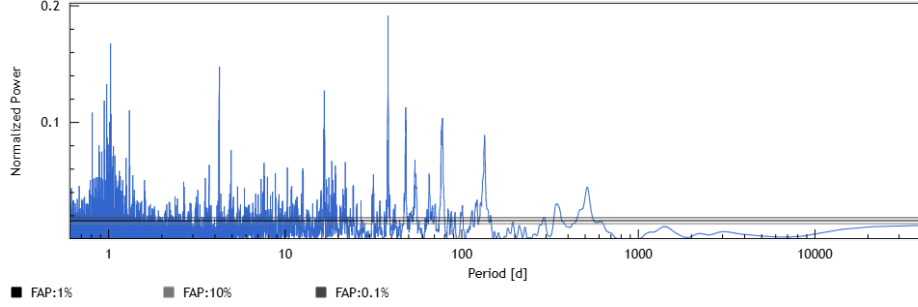


Figure 6: Periodogram - HD 115617

The Keplerian mass for the orbital time period 4.2149363 days is $5.399 M_{\oplus}$. Further, the Half-amplitude of the signal is measured to be 2.1468 m/s and the eccentricity, 0.0931.

2.2 Transit Method

Transit method measures the temporary dip in the star's brightness when a planet passes between its host star and the observer. The Kepler Space Telescope and the TESS (Transiting Exoplanet Survey Satellite) are two major space observatories that have used the transit method to discover thousands of exoplanets.

2.2.1 GJ 876

Getting the time period data from The Lick-Carnegie Exoplanet Survey Rivera et al. 2010 to be 1.937 days, and the photometric data from NASA Exoplanet archive IPAC Exoplanet Archive 2024, we get Figure 7.

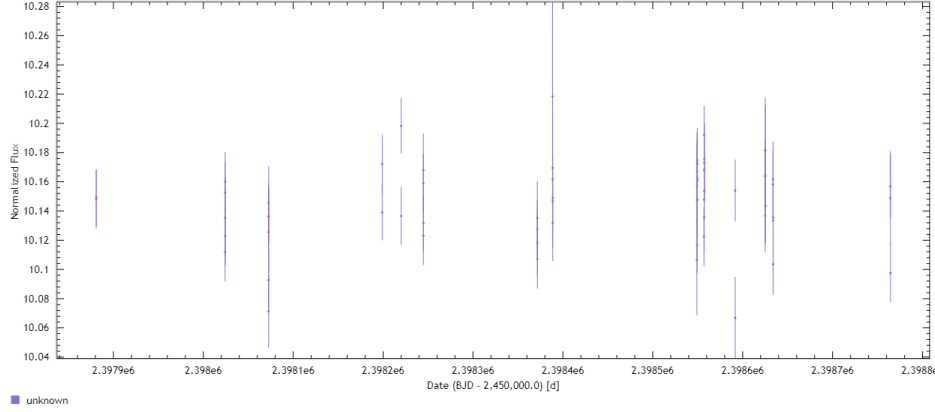


Figure 7: Photometry signal - GJ 876

The data do not show a transit signal but a rotational signal.

2.2.2 WASP 28

With time period data Anderson et al. 2015 to be 3.4088 days, and the photometric data from Vizier Données astronomiques de Strasbourg 2024, we use Dace to plot a transit curve. At 3.372 days, we get Figure 8.

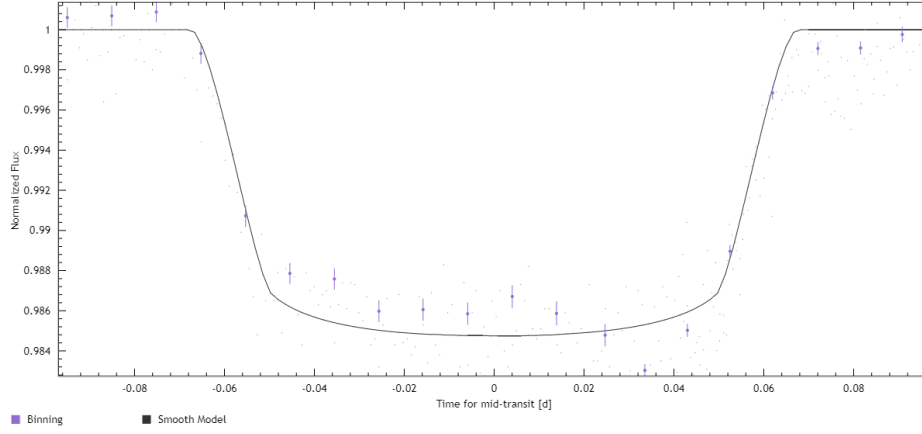


Figure 8: Transit Signal - WASP 28

Figure 8 fits perfectly with the model curve telling that a transit is happening.

2.2.3 BD +462629

After obtaining the time period data Howell et al. 2019 to be 1.76 days, and the photometric data from Vizier Données astronomiques de Strasbourg 2024, we get Figure 9 and Figure 10 showing the signal without phase folding and with model curve fitted.

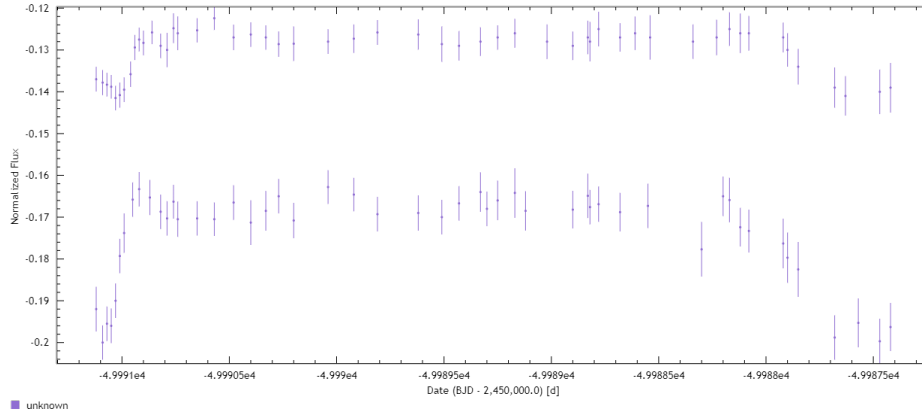


Figure 9: Photometry signal - BD +462629

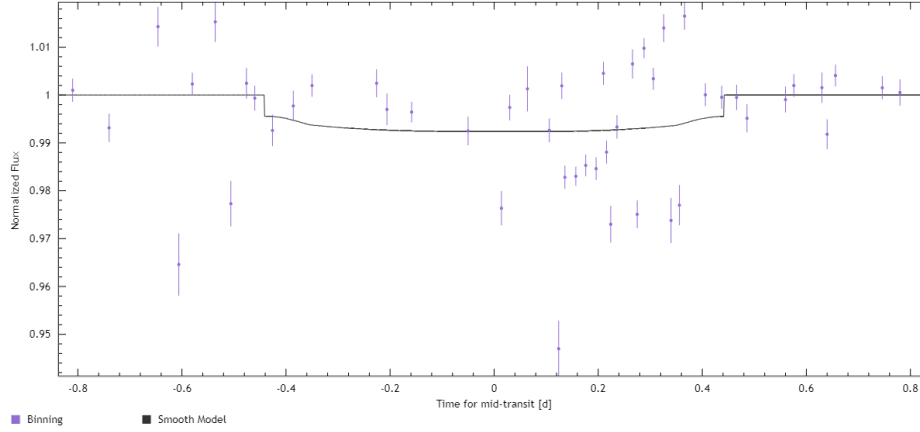


Figure 10: Transit model curve fitted image - BD +462629

From Figure 10 we can say that the data does not align with the model curve showing that no transit takes place.

3 Literature Review

3.1 GJ 551

Proxima Centauri, the closest star to the Sun, is part of the Alpha Centauri system and hosts at least one confirmed planet, Proxima Centauri b. Literature sources (Anglada-Escudé et al. 2016) report an orbital period of 11.186 days and a minimum mass of 1.27 Earth masses ($M_{\oplus}$) for Proxima Centauri b. The planet lies within the star's habitable zone.

3.2 HD 22049

Epsilon Eridani is a nearby K2V star located approximately 10.5 light-years away in the constellation Eridanus. Literature data indicate an

orbital period of 2502.1 days and a minimum mass of $273.09 M_{\oplus}$ (0.82 Jupiter masses) for its confirmed planet, Epsilon Eridani b (Tan et al. 2024, Hatzes et al. 2000). This system is notable for its debris disk, which suggests ongoing planetary formation or recent dynamical interactions.

3.3 HD 115617

The 61 Virginis system, situated 27.8 light-years away in the constellation Virgo, contains three confirmed planets, including super-Earths and Neptune-mass bodies. The innermost planet has a published orbital period of 4.2150 days and a minimum mass of $5.1 M_{\oplus}$ (Vogt et al. 2010). The system's planets are located close to

their host star, making it an ideal target for studying the formation and evolution of compact planetary systems.

3.4 GJ 876

Located 15 light-years away in the constellation Aquarius, GJ 876 is a multi-planet system with four confirmed planets, including gas giants and terrestrial bodies. Literature values for the shortest-period planet indicate an orbital period of 1.93781 days (Rivera et al. 2010). This system is known for its resonant planetary orbits, which provide critical insights into the migration and interaction of planets.

3.5 WASP 28

WASP-28, situated approximately 200 parsecs away, hosts a

confirmed hot Jupiter, WASP-28b. Literature sources report an orbital period of 3.40883 days (Anderson et al. 2015). This inflated Jupiter-mass planet is in a near-aligned orbit around its host star, making it an excellent candidate for atmospheric characterization via transmission spectroscopy.

3.6 BD+462629

BD+462629, also known as Kepler-13AB, is a visually close binary star system located approximately 520 parsecs from Earth. It is host to the exoplanet Kepler-13b, one of the most massive known hot Jupiters. Literature data indicate that Kepler-13Ab has an orbital period of 1.77 days and a mass of approximately 6 Jupiter masses (Howell et al. 2019).

4 Discussion

Table 1 and Table 2 compares the data observed in this study and the data from published papers.

Targets	GJ 876	WASP 28	BD+462629
Time Period used (days)	1.944724	3.37252759	1.77
Time period from source (days)	1.937781	3.40883	1.76
Signal type observed	Rotational	Transit	Rotational
Signal Type from papers	Transit	Transit	Transit

Table 1: Comparison Table for Transit Method Targets

Targets	GJ 551	HD 22049	HD 115617
Observed Half-Amplitude (m/s)	1.5658	12.0173	2.1468
Half-Amplitude from papers (m/s)	1.38	19.0	2.12
Observed Eccentricity	0.3389	0.04080	0.0931
Eccentricity from papers	0.35	0.608	0.12
Observed minimum Mass $M_{\oplus}$	5.153	259.965	5.399
Minimum mass from papers $M_{\oplus}$	1.27	273.09	5.1
Observed Period (days)	11.191643	2649.791	4.2149363
Period from papers (days)	11.186	2502.1	4.2150

Table 2: Comparison between Published papers and RV method data

4.1 Published Papers Outlook

The minimum mass of GJ 551b - the probable cause of the fluctuation in radial velocity at 11.2 days - is 1.3 Earth mass $M_{\oplus}$ way less than the minimum mass predicted by the keplerian fit. This could be due to the conservative estimate of inclination used in the published paper Anglada-Escudé et al. 2016. Kepler fit always uses 90° as the inclination hence giving a larger minimal mass.

In addition, the discrepancies in HD 22049 could also be due to application of modern methods or refinements in orbital fitting that weren't available for the earlier work. New data since the publication of Hatzes et al. (2000) Hatzes et al. 2000 provide a better understanding of the system's orbital parameters, leading to refined results. The precision

of the radial velocity measurements could also improve with time, leading to updated parameters.

For HD 115617 (61 Virginis), the observed half-amplitude of 2.1468 m/s agrees closely with the literature value of 2.12 m/s, indicating consistent results. The observed eccentricity is 0.0931, slightly lower than the literature value of 0.12, likely due to differences in noise modeling during data analysis.

For GJ 876, the observed period is 1.944724 days compared to 1.937781 days from the source. This slight difference can result from discrepancies in the dataset used for orbital fits or differences in assumptions about stellar activity. Observed signal type is rotational, whereas literature mentions it as transit, suggesting possible misclassification in signal interpretation or errors during observations.

For WASP 28, the observed pe-

riod is 3.37252759 days, slightly lower than the literature period of 3.40883 days. This difference may arise due to variations in the treatment of transit timing variations (TTVs). Both observed and literature data agree on the signal type as transit.

For BD+462629 (Kepler-13AB), the observed orbital period is 1.77 days, closely matching the literature value of 1.76 days. The observed signal type is rotational, differing from the transit classification in literature. This discrepancy may stem from observational limitations or misclassification due to binary system complexities, as Kepler-13AB is a binary star system with unique dynamical properties that might affect data interpretation

4.2 Exo-planets properties

Exoplanet exhibit characteristics not observed in the Solar System, such as extremely eccentric orbits, high densities (super-Earths), or very low densities (puffy gas giants) because of the diversity in their orbital and physical properties, formation histories and environmental factors. If the exoplanets orbit their stars closely, strong gravitational interactions produce pronounced radial velocity variations and transit depths.

Despite these differences, the de-

tection methods for exoplanets often rely on principles that are analogous to our understanding of Solar System planets. The radial velocity method relies on Kepler’s law of orbital motion, created with solar system as base and the transit method mirrors the eclipse concept, since the fundamental physics of gravity and orbital mechanics applies universally.

Figure 11 is a plot of exoplanets discovered till date (data from NASA Exoplanet Archive NASA Exoplanet Archive 2024) and their properties. It reveals several patterns: most detected exoplanets cluster around shorter orbital periods and larger masses due to the biases of current detection techniques. Super-Earths and hot Jupiters dominate the data, while Earth-like planets remain underrepresented due to the difficulty of detecting small, distant planets.

This plot also reveals incompleteness in measurement techniques. Transit and radial velocity methods are most sensitive to planets with short orbital periods and high masses. Conversely, long-period and small planets, akin to Neptune or Earth, are underrepresented due to limited observational timescales and the smaller signals they produce. Direct imaging, on the other hand, favors distant, massive planets, further highlighting these biases.

Host stars for exoplanets differ significantly from the Sun in many

cases. Many are smaller, cooler M-dwarfs, where the habitable zone lies closer to the star, enabling easier detection of potentially Earth-like planets. Some host stars are also younger or more metal-rich than the Sun, which influences planet formation, leading to a higher likelihood of massive planets.

Finally, the number and types of exoplanets change notably with host star properties. Stars with higher

metallicity tend to host more massive planets, as metallicity correlates with the availability of solid material in the protoplanetary disk. Stellar mass also plays a role: low-mass stars frequently host compact multi-planet systems, while higher-mass stars more often harbor gas giants. These relationships illustrate how stellar properties shape planetary systems, reinforcing the diversity observed in exoplanet discoveries.

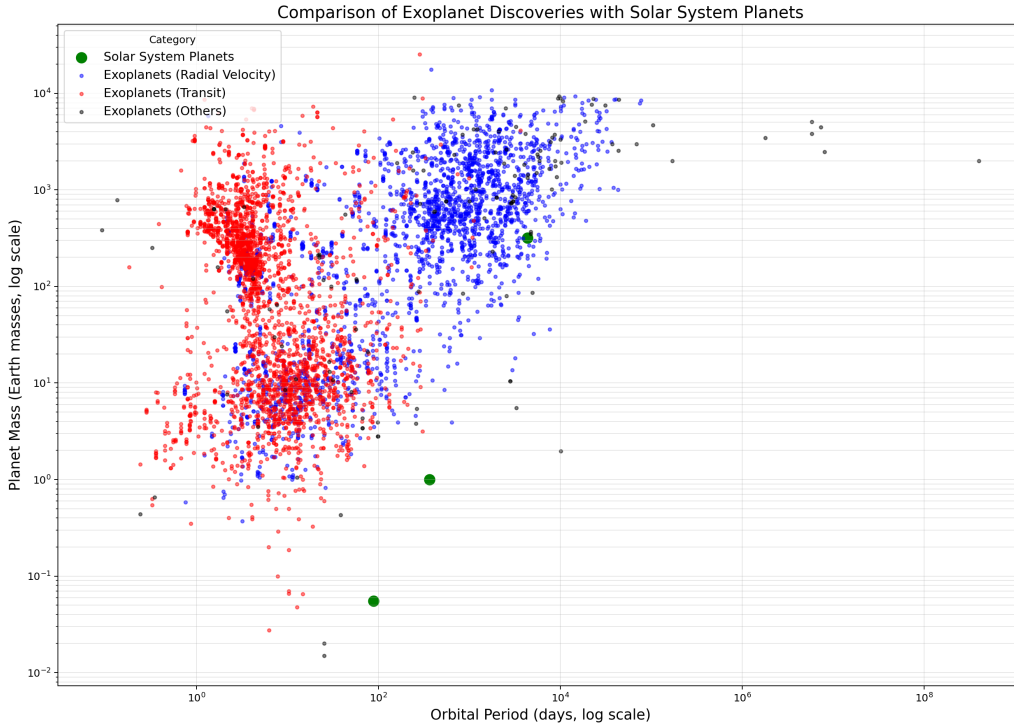


Figure 11: Relation between Exoplanet properties and detection

5 Conclusion

The analysis of six stars using radial velocity and transit methods has provided significant insights into exoplanet detection and characterization. Three stars—GJ 551, HD 22049, and HD 115617—revealed periodic radial velocity signals indicative of planetary companions, with estimated minimum masses and orbital parameters consistent with, yet distinct from, existing literature. Transit analysis confirmed the presence of a planetary transit for WASP 28, while no such signals were observed for GJ 876 and BD+462629.

The comparison of observed results with published data highlighted notable discrepancies in mass and eccentricity estimates, attributable to differences in datasets, methodologies, and model assumptions. These findings emphasize the necessity of integrating advanced observational technologies and multi-method approaches to refine our understanding of exoplanetary systems.

This case study underscores the evolving precision of exoplanet detection techniques, paving the way for future research into the formation, dynamics, and habitability of planets beyond our Solar System.

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